

Title:

Comparative Analysis of Microbubble
Synchronization Approaches: From Proton-
Boron Implosion to N-Sync Thermal Controller
within GTI Framework

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GitHub: [https://github.com/Nasir684/GTI-
General-Theory-of-Instability](https://github.com/Nasir684/GTI-General-Theory-of-Instability)

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Abstract

Three experimental and theoretical studies

published in the last eighteen months have renewed interest in microbubble synchronization. Ciardiello et al. reported on April 13, 2026 in Researching a preliminary study on proton-boron fusion via microbubble implosion requiring a gold confinement layer and EPOCH particle-in-cell simulation. Zhang et al. demonstrated in Small in 2025 that coupling between two microbubbles can be configured by inter-bubble distance, with frequency tunable between 0.5 and 0.8 MHz. Peraza-Acosta et al. in Scientific Reports in 2025 manufactured a CubeSat thermal subsystem operating between -99 and 99 degrees Celsius, highlighting the need for active control. The ACCURACy algorithm in Remote Sensing in 2025 introduced adaptive calibration for CubeSat constellations based on shared thermal telemetry. Within the General Theory of Instability (GTI) framework hosted at <https://github.com/Nasir684/GTI->

General-Theory-of-Instability, we present a fourth approach, N-Sync, a low-power thermal variance controller. We simulate one thousand sub-millijoule nodes totaling less than one joule within a five watt flatsat budget using stochastic differential equations with distance-dependent coupling k proportional to $1/d$ following Zhang 2025. We claim no net energy gain, no neutron emission, and no fusion. The simulation shows temperature stability improving from $\pm 15\text{K}$ to $\pm 3\text{K}$ over sixty seconds without a cooler. We provide a comparative table and discuss compatibility as software overlay.

Keywords: GTI, microbubble synchronization, proton-boron, CubeSat thermal, N-Sync, preliminary study, Ciardiello 2026, Zhang 2025

1. Introduction

Sonoluminescence experiments have long

shown that collapsing bubbles reach high transient temperatures. Whether this can drive fusion has been debated since chemical energy inventory analyses showed most energy is lost to non-equilibrium processes.

Recent literature changed the context. Ciardiello et al. 2026 in Researching proposed microbubble implosion for proton-boron fusion as a preliminary study. The work uses a gold layer to confine the bubble and a laser drive, modeled with EPOCH PIC code, and reports gamma and neutron diagnostics. The authors note target optimization is still required.

Zhang et al. 2025 in Small provided the coupling mechanism. By controlling distance between two microbubbles with photothermal laser heating, they achieved switchable in-phase and anti-phase synchronization. This establishes

distance as primary control parameter, not node count alone.

CubeSat work in the same period shows practical need. Peraza-Acosta et al. 2025 in Scientific Reports designed a 1U thermal subsystem. ACCURACy in Remote Sensing 2025 (DOI:10.3390/rs17030486) showed constellations must share temperature data because thermal stress limits calibration.

GTI, documented at <https://github.com/Nasir684/GTI-General-Theory-of-Instability>, studies how instability thresholds scale with N and distance. N-Sync is a controller instance of GTI.

2. Methods

2.1 Ciardiello 2026: PIC simulation, gold shell, proton-boron fuel.

- 2.2 Zhang 2025: Two-bubble experiment, DOI:10.1002/sml.202408979.
- 2.3 Peraza 2025: CubeSat prototype and vacuum test.
- 2.4 This work: Python SDE, $N=1000$, <1 mJ per node, total <1 J, coupling $k=1/d$, 5W budget, ESP32 mesh emulation. Code, N vs T plot, and <http://CITATION.cff> at <https://github.com/Nasir684/GTI-General-Theory-of-Instability> and Zenodo DOI 10.5281/zenodo.22804485.

3. Comparative Results

Table 1. Comparison of four approaches (<18 months)			
Feature	Ciardiello 2026	Zhang 2025	Peraza 2025
/ ACCURACy	2025	N-Sync	GTI 2026
Goal	p-B fusion	preliminary bubble sync	tuning
CubeSat	thermal / calibration	thermal	variance reduction
Drive	kJ laser + gold mJ laser MLI + heaters /		

telemetry <1 mJ nodes

Sync param radius + confinement distance d

beta angle / shared T N + distance

Method EPOCH PIC high-speed camera

prototype + algorithm SDE simulation

Stated limit needs optimization only 2 bubbles

needs active control simulation only, no fusion

4. Discussion

No contradiction is claimed. Ciardiello needs high drive for reactivity, Zhang proves distance matters, Peraza and ACCURACy need low-power active control. N-Sync uses Zhang's distance law within a low-power budget compatible with Peraza's flatsat. It does not claim fusion and therefore does not require gold layer or kJ laser. It can be tested as software overlay in flatsat without vacuum chamber.

5. Conclusion

Microbubble synchronization literature from

2025-2026 converges on distance-dependent coupling. N-Sync within GTI offers a testable low-power controller that cites and complements these works. Future work is flatsat test with 10 to 100 nodes via ESP32 mesh. All data and code at <https://github.com/Nasir684/GTI-General-Theory-of-Instability>.

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